CO2xN: Porewater Methane

Month Year Samples

2026-04-10

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##Run Information

```
cat("Run Information: Melanie Giessner ") #lets you know what section you're in
```

```
## Run Information: Melanie Giessner
```

```
#set the run date & user name
sample_year <- "2026"
sample_month <- "March & April"
user <- "Melanie Giessner"
#identify the files you want to read in
#read in as a list to accommodate multiple runs in a month

GHG_files <- c("Raw Data/20260406_GENX_GHGdata.csv" )


# Define the file path for QAQC log file - NO Need to change just check year
log_path <- "Processed Data/LTREB_2M_ShimadzuGC_QAQC_2025.csv"


# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/GCReW_GENX_Porewater_CH4_20260406.csv"

#Gas in Water Calculations
# "TempC" column value in Celsius (equilibration temperature) and volume of gas (in liters) in syringe
TempC = 25 #Equilibration Temperature (Celsius)

volume_g = 0.015 #volume of gas (in liters) in syringe when equilibrized
volume_w = 0.015 #volume of water (in liters) in syringe when equilibrized (should be equal to volume_g

#record any notes about the run or anything other info here:
run_notes <- ""

#Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"

cat(run_notes)
```

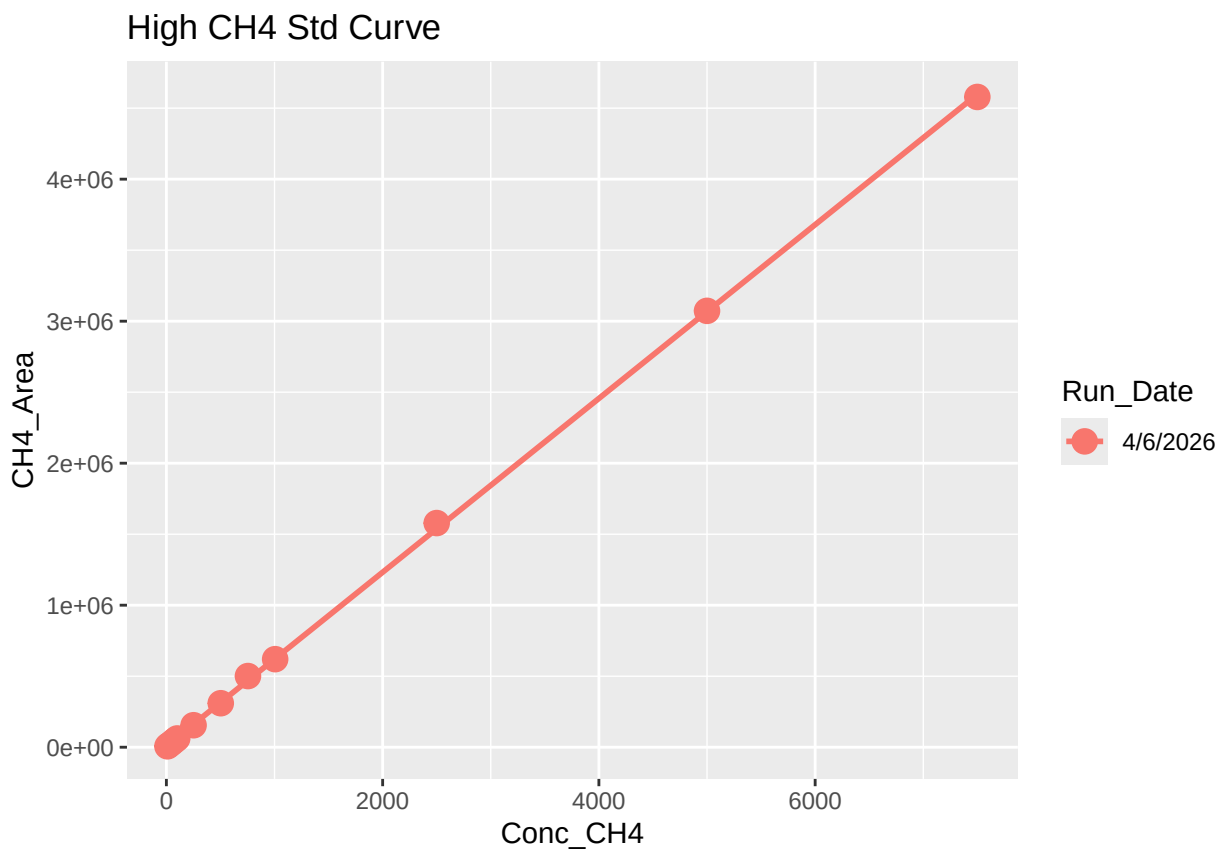
##Set Up

##Read in metadata and create similar sample IDs for matching to samples

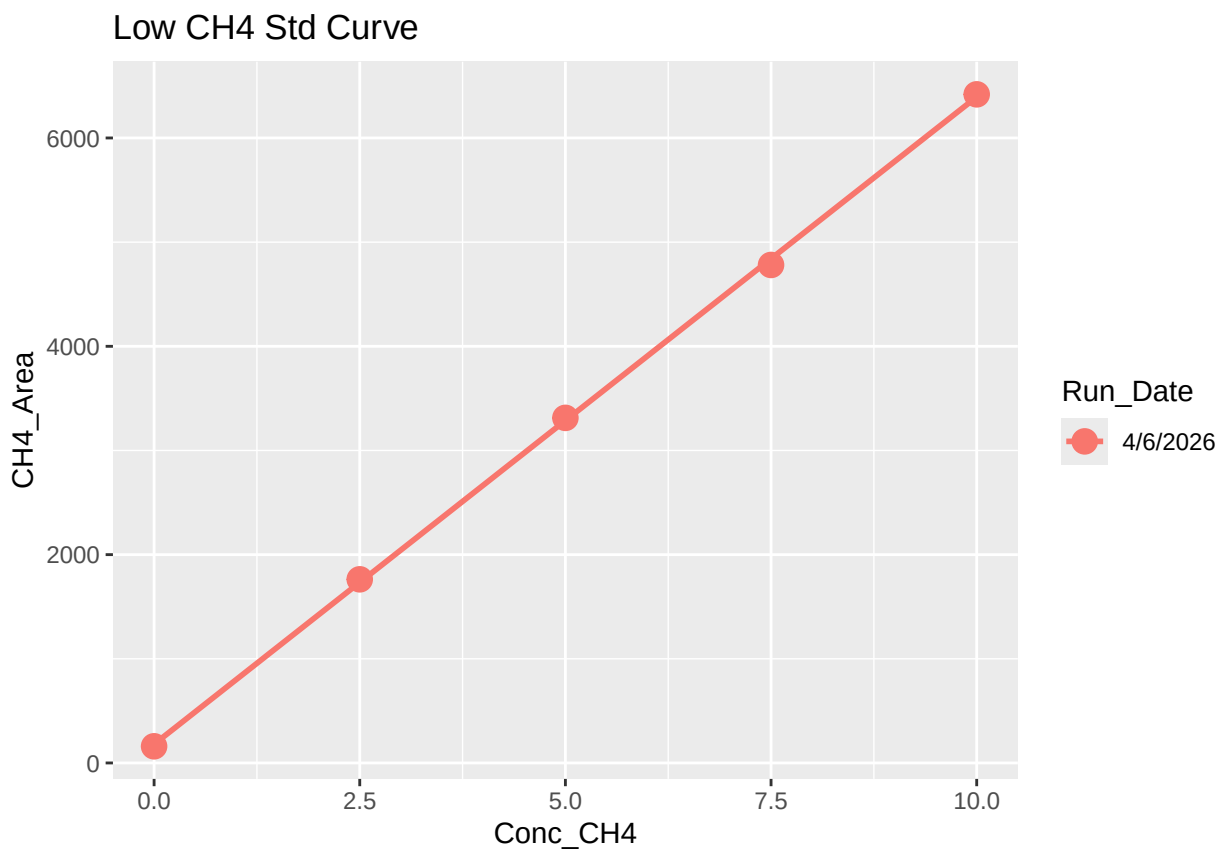
1 Read in data files

2 Assess standard curves

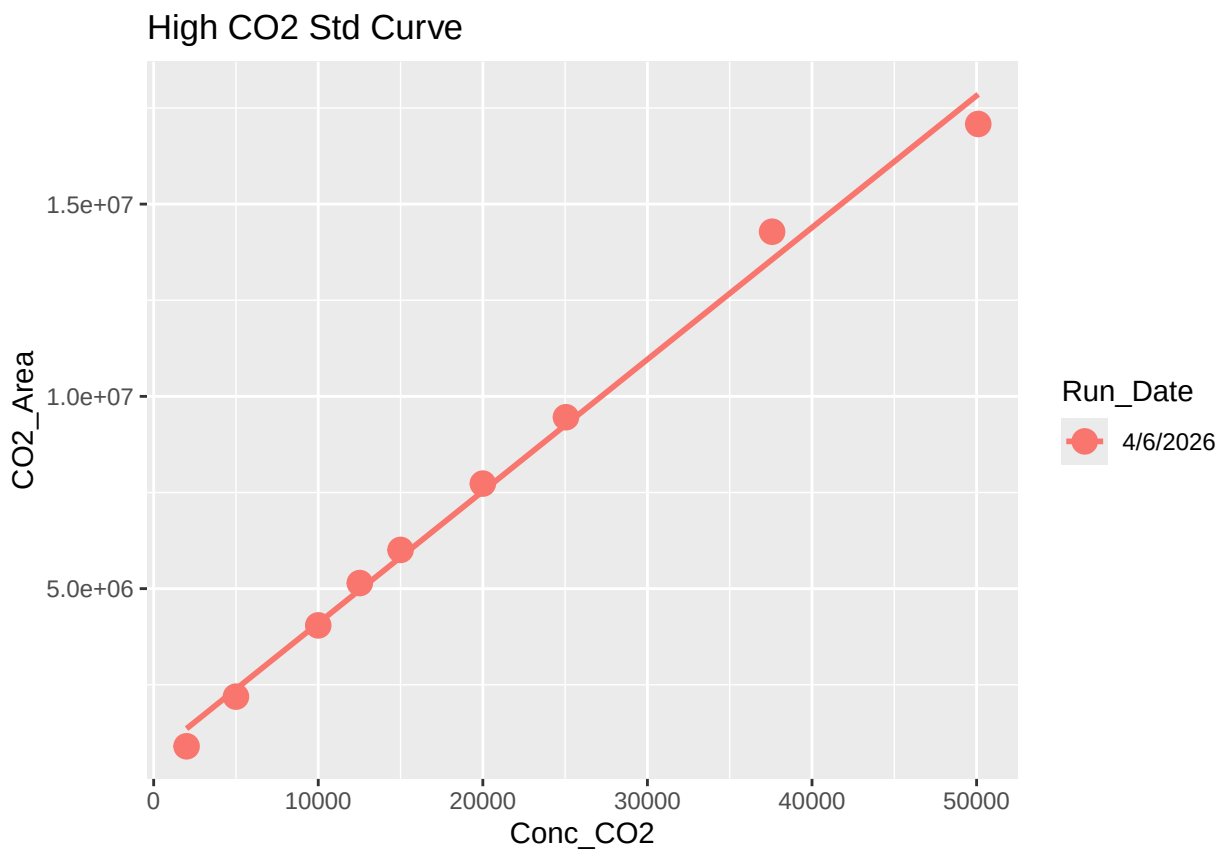
```
## `geom_smooth()` using formula = 'y ~ x'
```



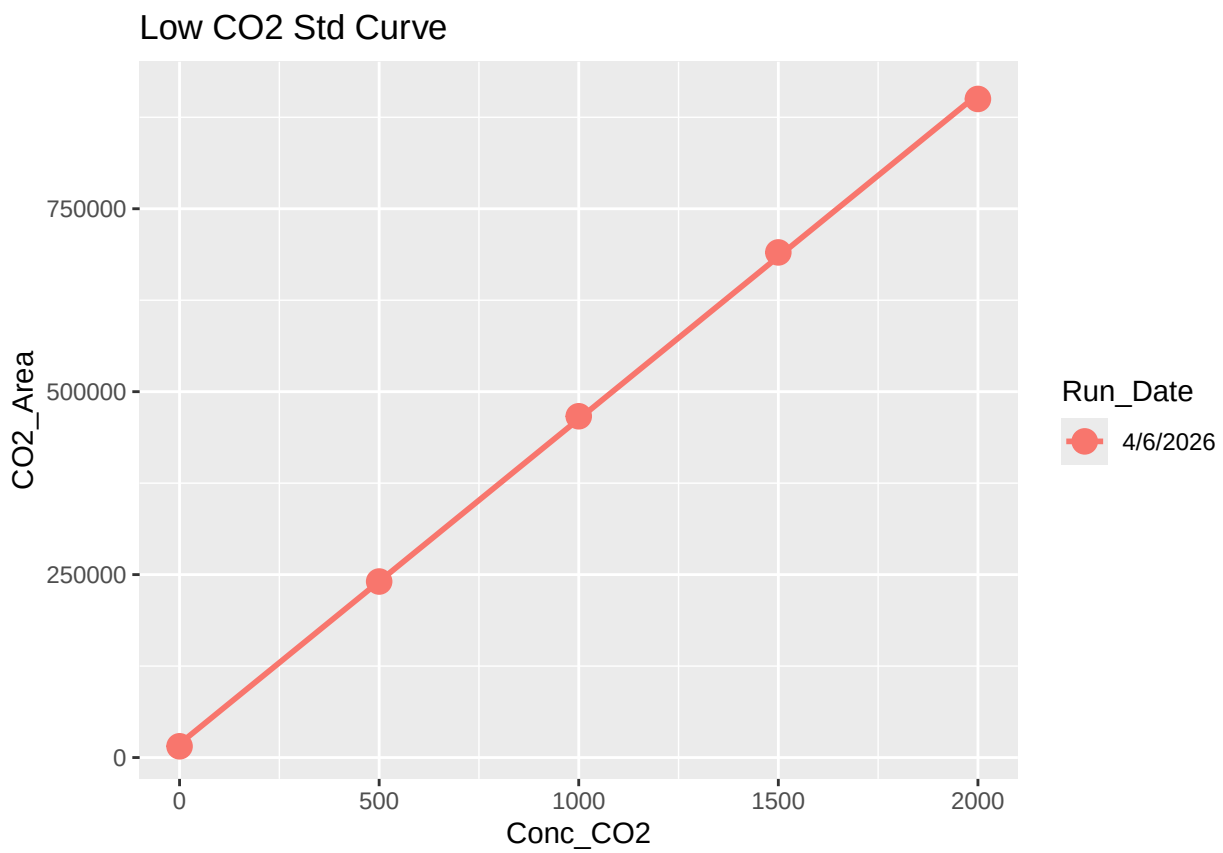

```
## `geom_smooth()` using formula = 'y ~ x'
```



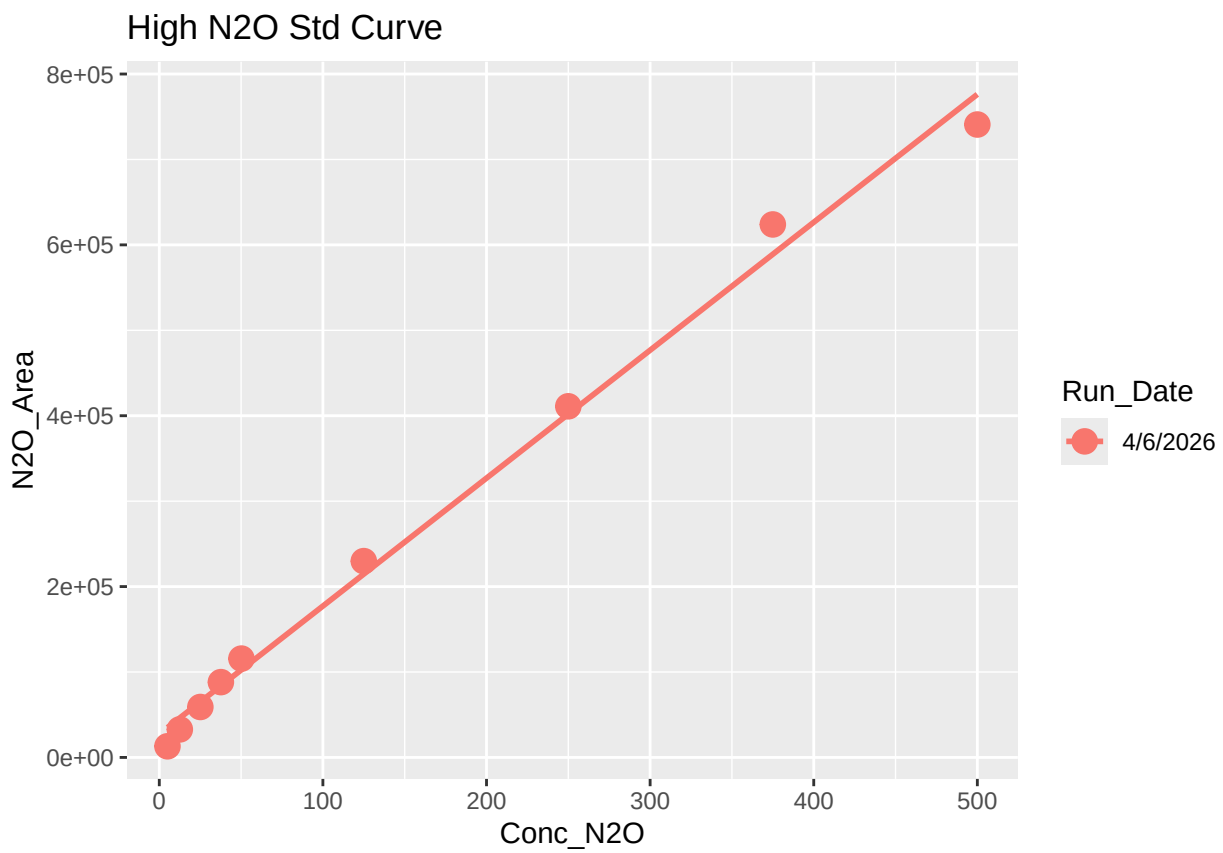

```
## Joining with `by = join_by(Run_Date)`

## [1] "High_R2 is Good-4/6/2026"

## [1] "Low_R2 is Good-4/6/2026"

## `geom_smooth()` using formula = 'y ~ x'
```



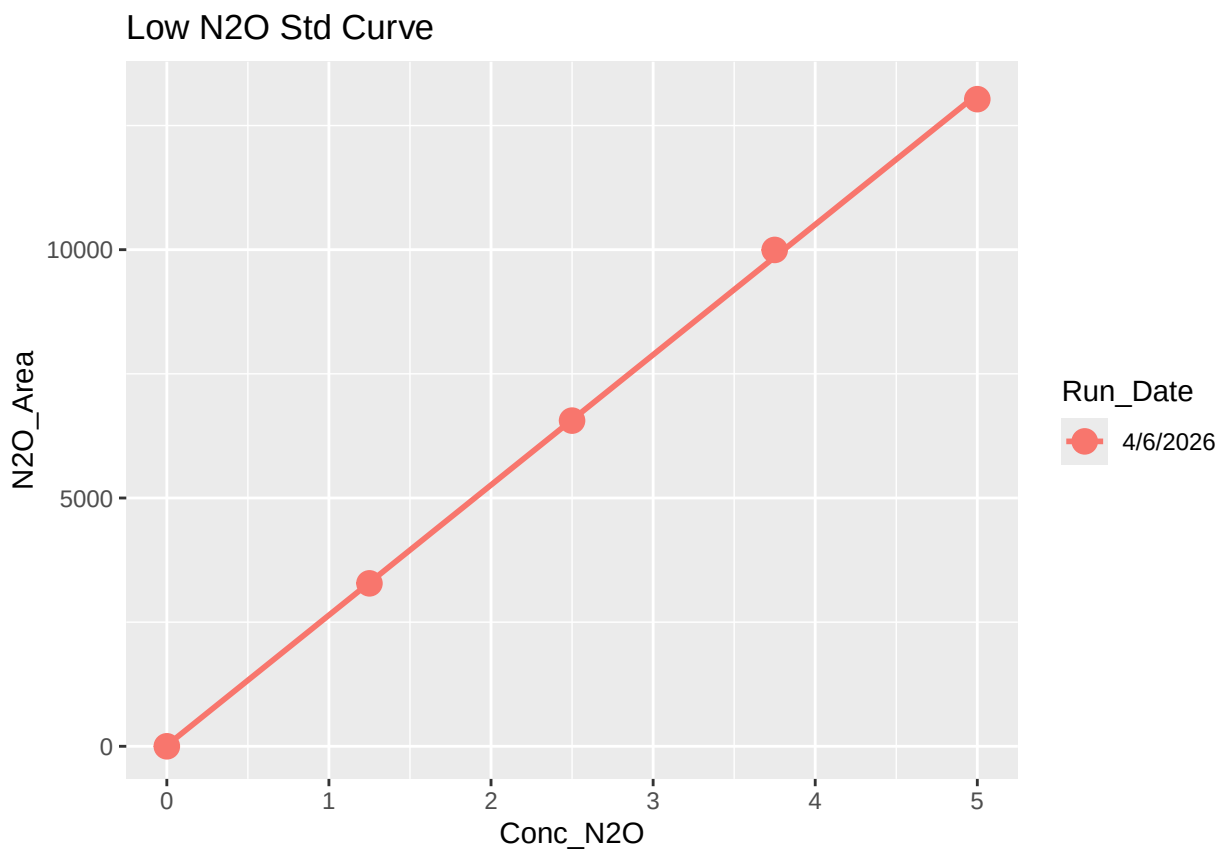

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
## Joining with `by = join_by(Run_Date)`

## [1] "High_R2 is Good-4/6/2026"

## [1] "Low_R2 is Good-4/6/2026"

## `geom_smooth()` using formula = 'y ~ x'
```




```
## `geom_smooth()` using formula = 'y ~ x'
```



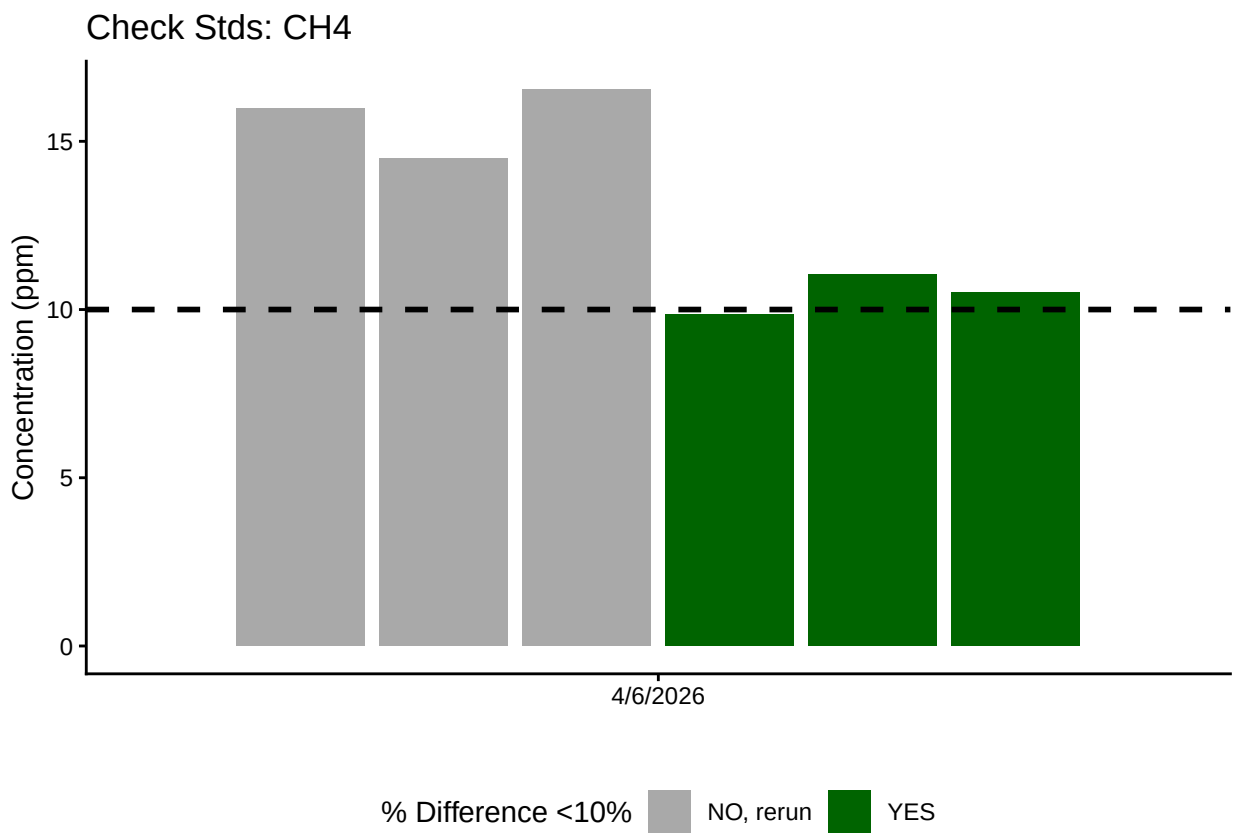

```
## Joining with `by = join_by(Run_Date)`

## [1] "High_R2 is Good-4/6/2026"

## [1] "Low_R2 is Good-4/6/2026"
```

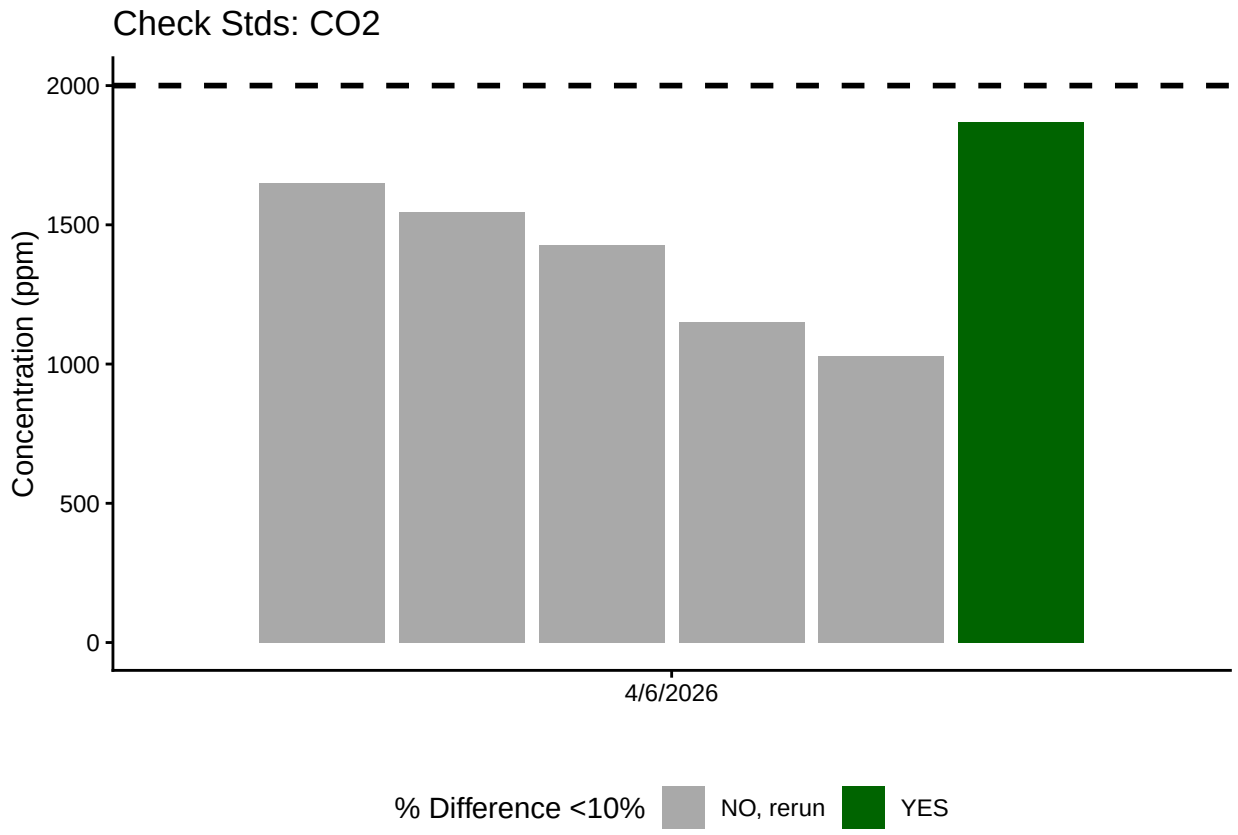
```
#add new standard curve data to log
# Slope_CH4<- Slope_CH4 %>% rename(CH4_Curve=Curve,CH4)
#
# log <- read.csv(log_path)
# head(log)
# log <- log[, -c(1)]
# log <- rbind(log, Slope_CH4)
#
# #remove possible duplicates from multiple QAQC attempts
#
# unique_log <- log %>% distinct(.keep_all = TRUE)
#
# Slopes_chk <- ggplot(log, aes(x = Run_Date, y = Slope, col = CH4_Curve)) +
#   geom_point(size = 4) +
#   ylim(400, 800) +
#   theme_bw()
#
# Slopes_chk

#write.csv(unique_log, log_path)
```

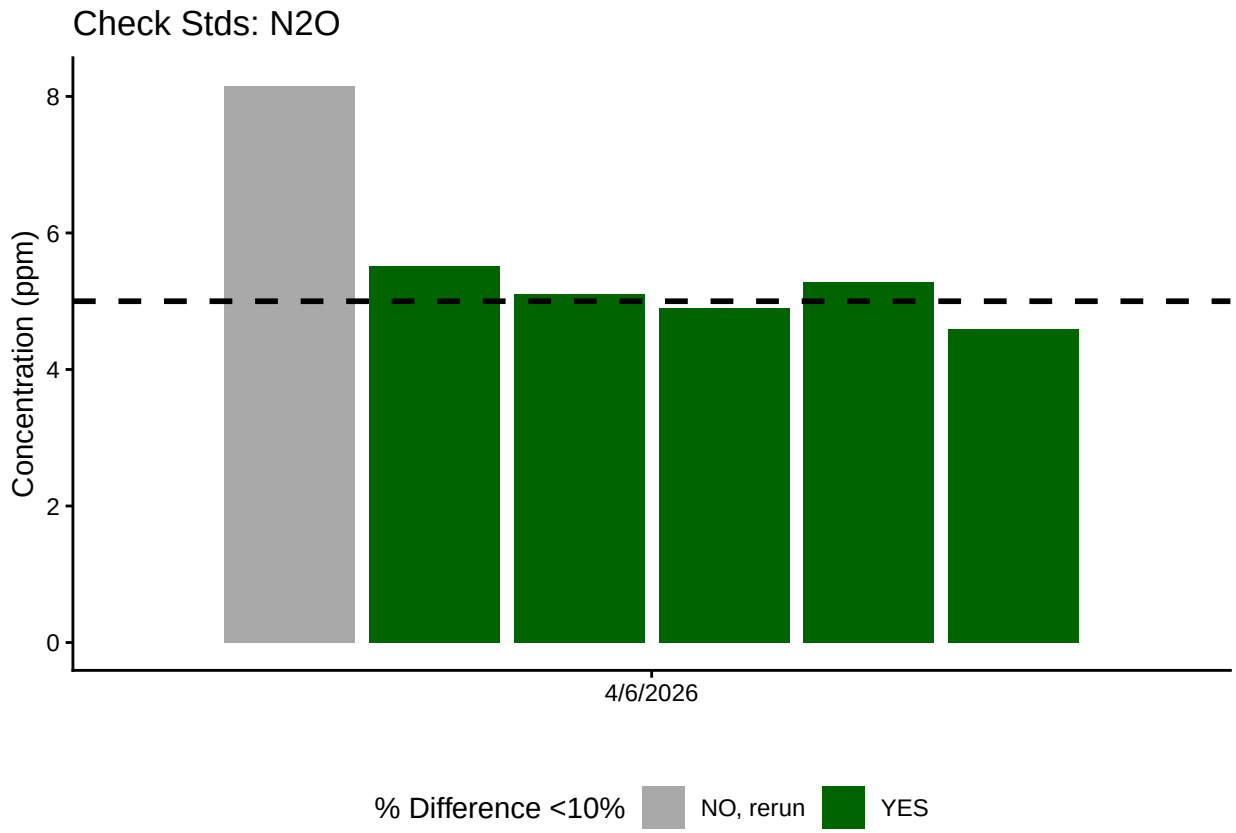
2.1 Now calculate the CH4 in ppm

3 Aanlayze the Check Standards

```
## <60% of CH4 Check Standards are within range of expected concentration - REASSESS

## <60% of CO2 Check Standards are within range of expected concentration - REASSESS

## >60% of N2O Check Standards are within range of expected concentration - PROCEED
```

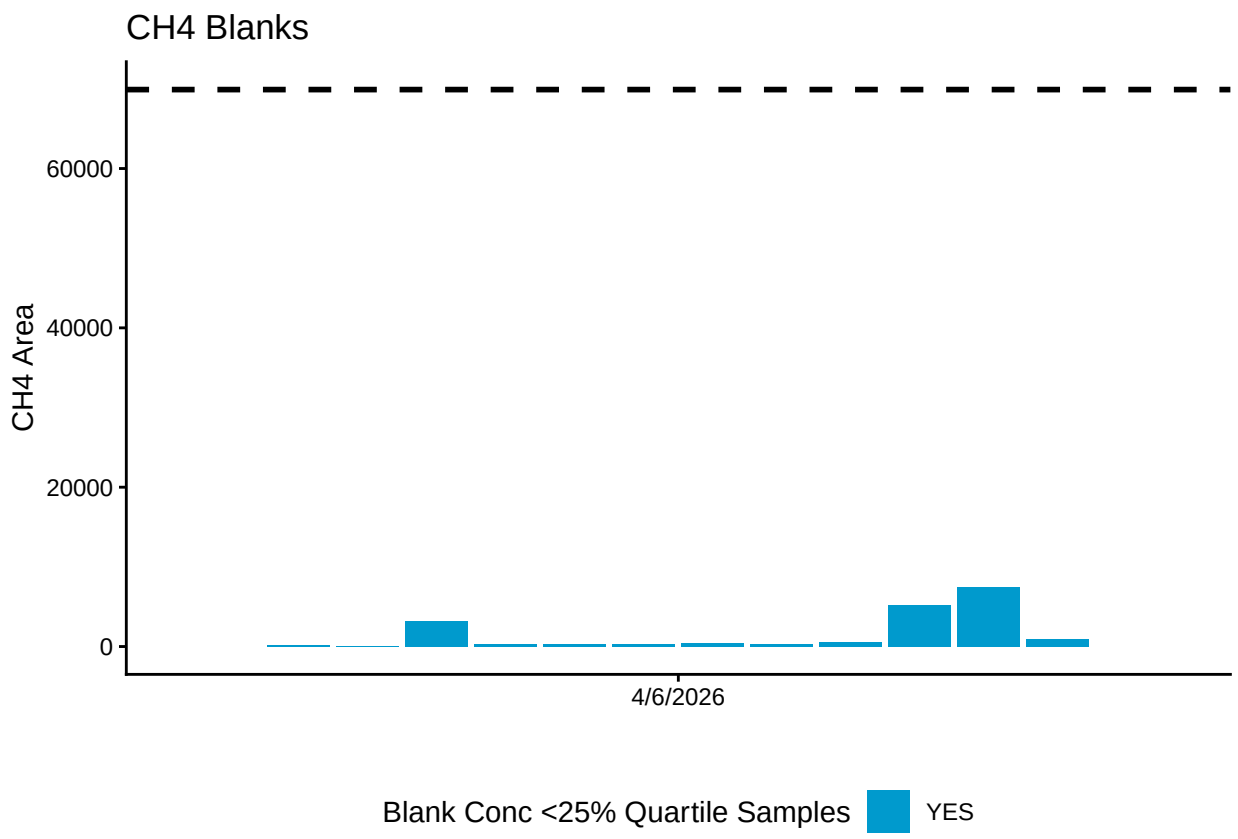


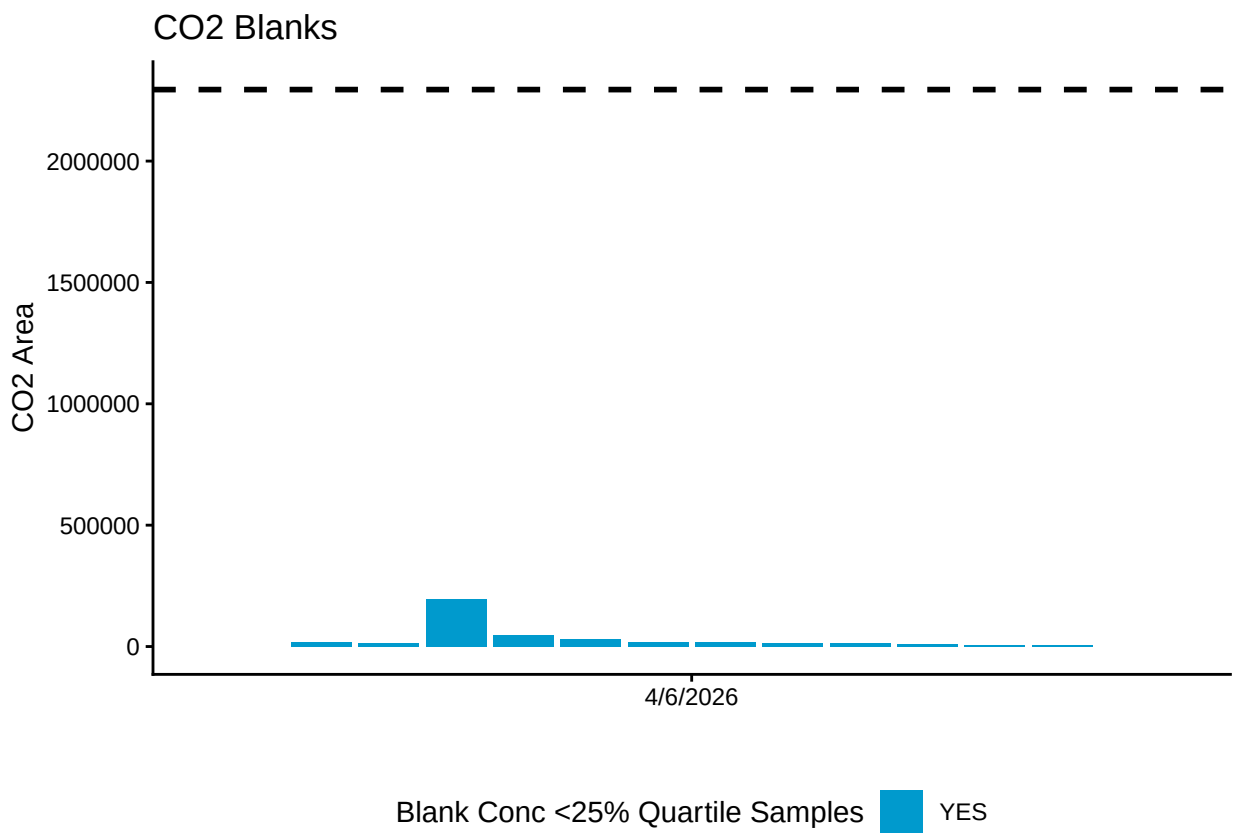

Check Stds: CO2
Concentration (ppm)
2000
1500
1000
500
0
4/6/2026
% Difference <10%
NO, rerun
YES



4 Analyze the Blanks

```
## Assess Blanks

## >60% of CH4 Blanks are below the lower 25% quartile of samples - PROCEED

## >60% of CO2 Blanks are below the lower 25% quartile of samples - PROCEED

## >60% of N2O Blanks are below the lower 25% quartile of samples - PROCEED
```


CH4 Blanks
CH4 Area
60000
40000
20000
0
4/6/2026
Blank Conc <25% Quartile Samples
YES


CO2 Blanks
CO2 Area
2000000
1500000
1000000
500000
0
4/6/2026
Blank Conc <25% Quartile Samples
YES

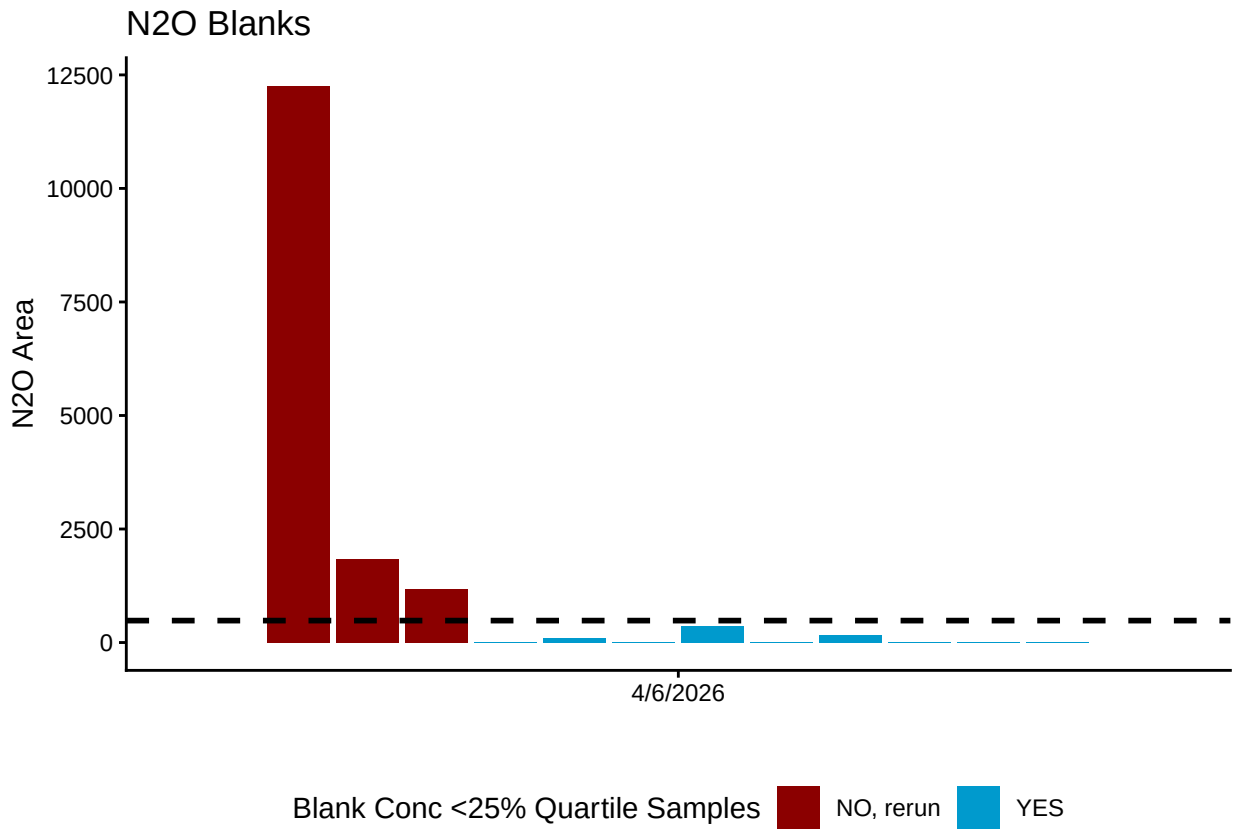
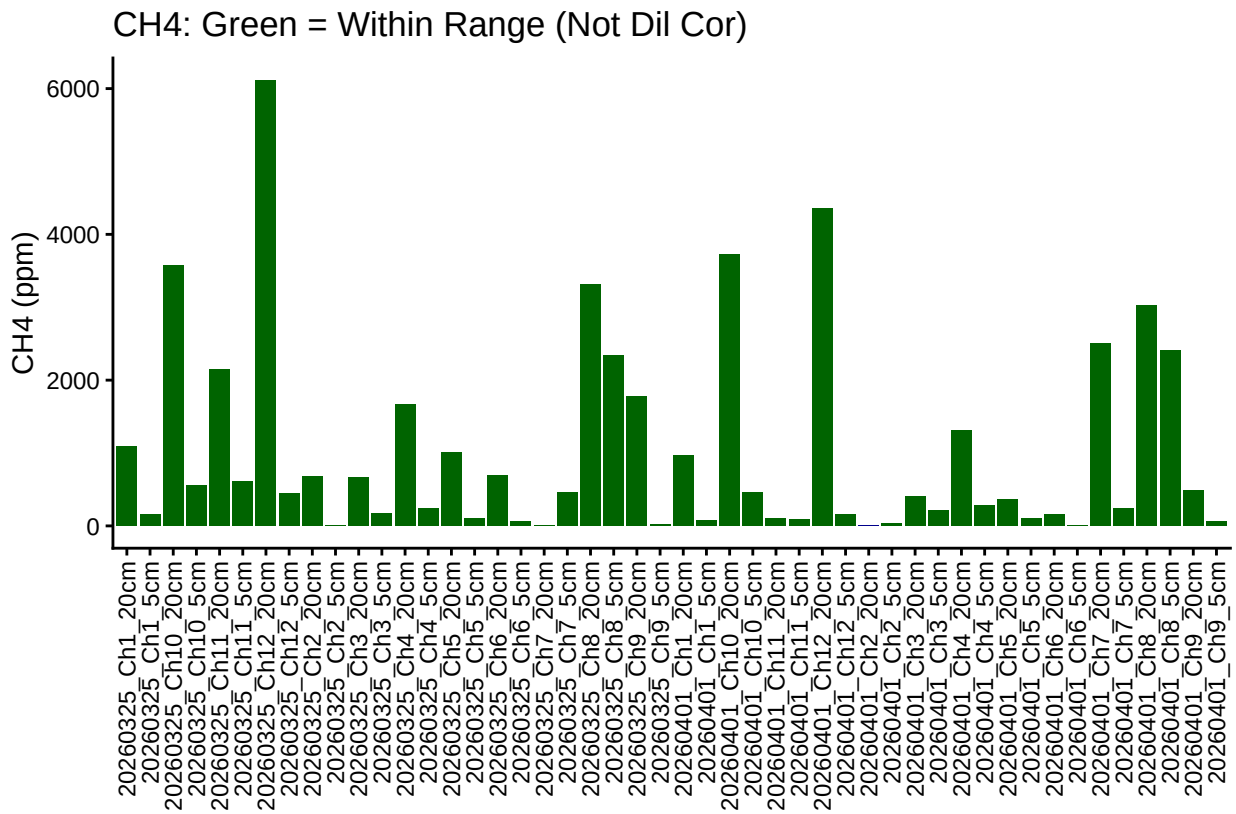
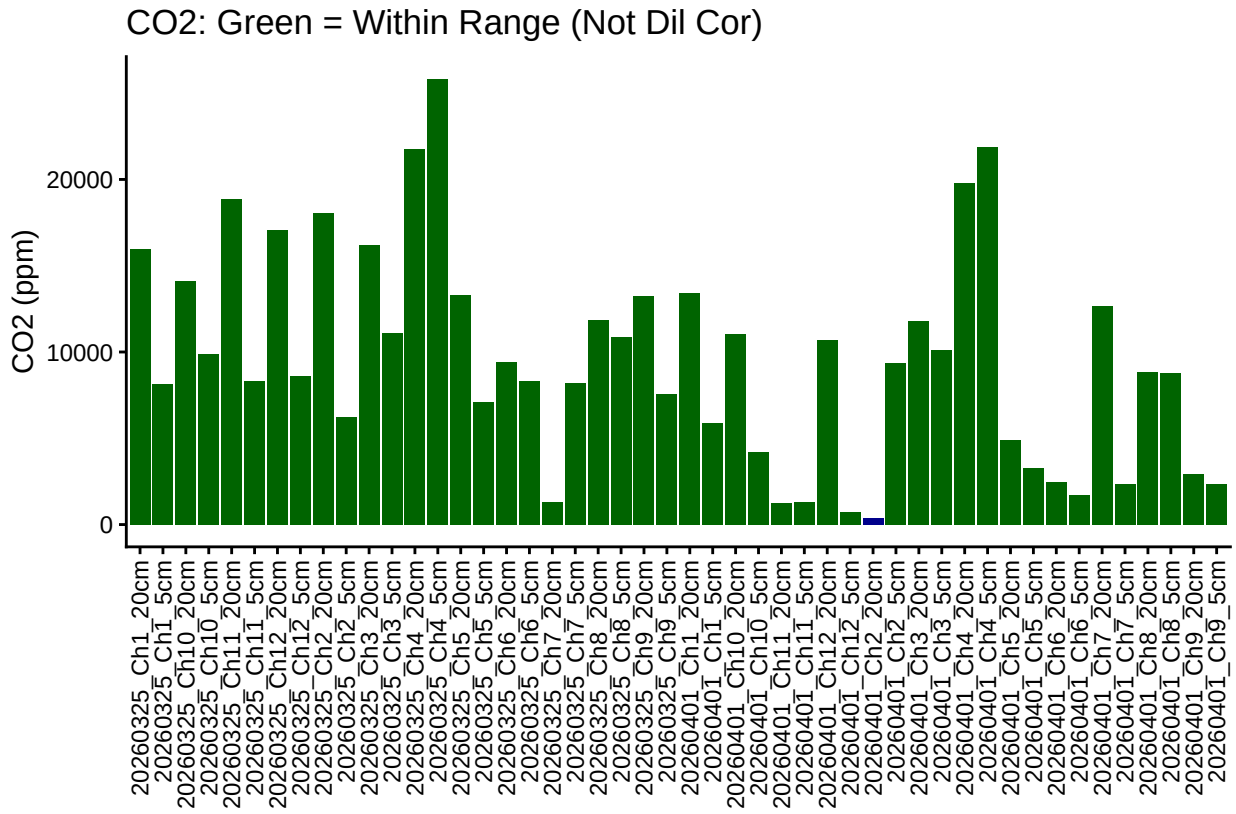


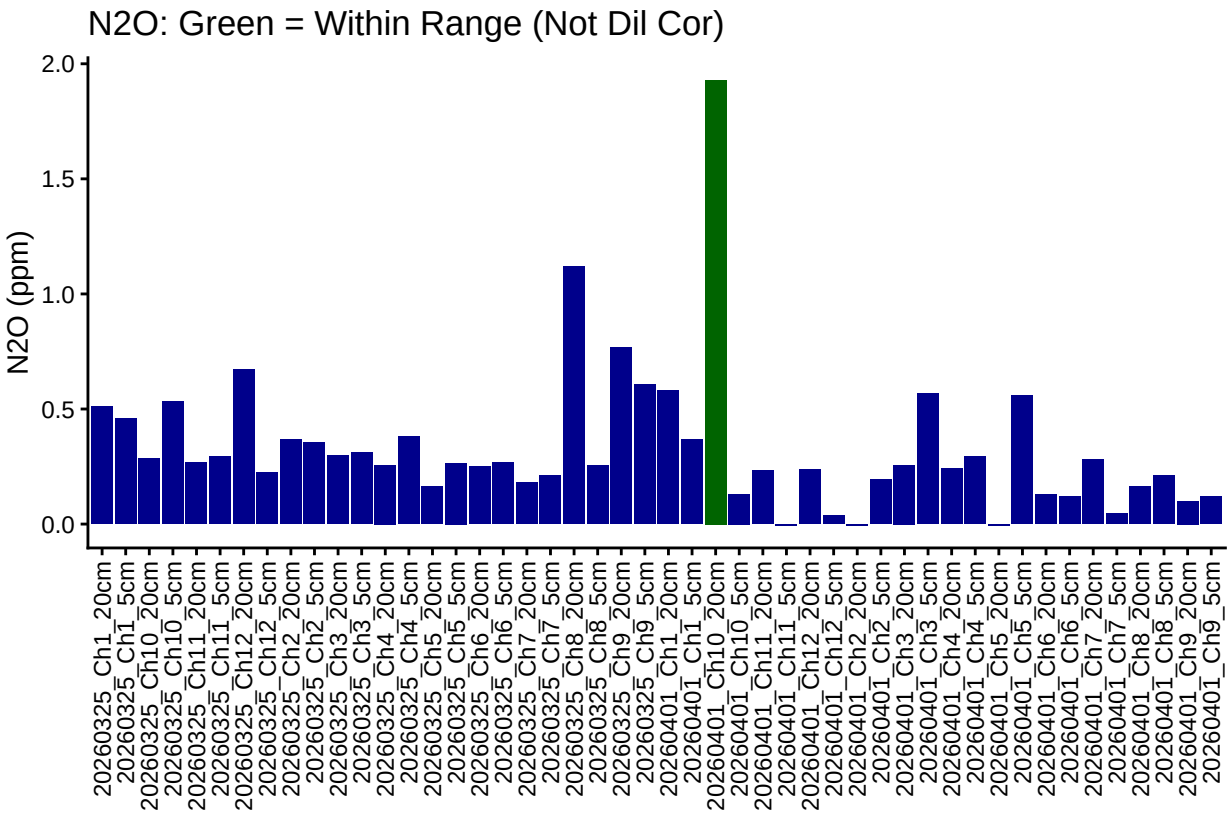
Table 1: Mean Area of Blanks

Test	Blank_Mean_Area
CH4	1623.000
CO2	33500.083
N2O	1324.583

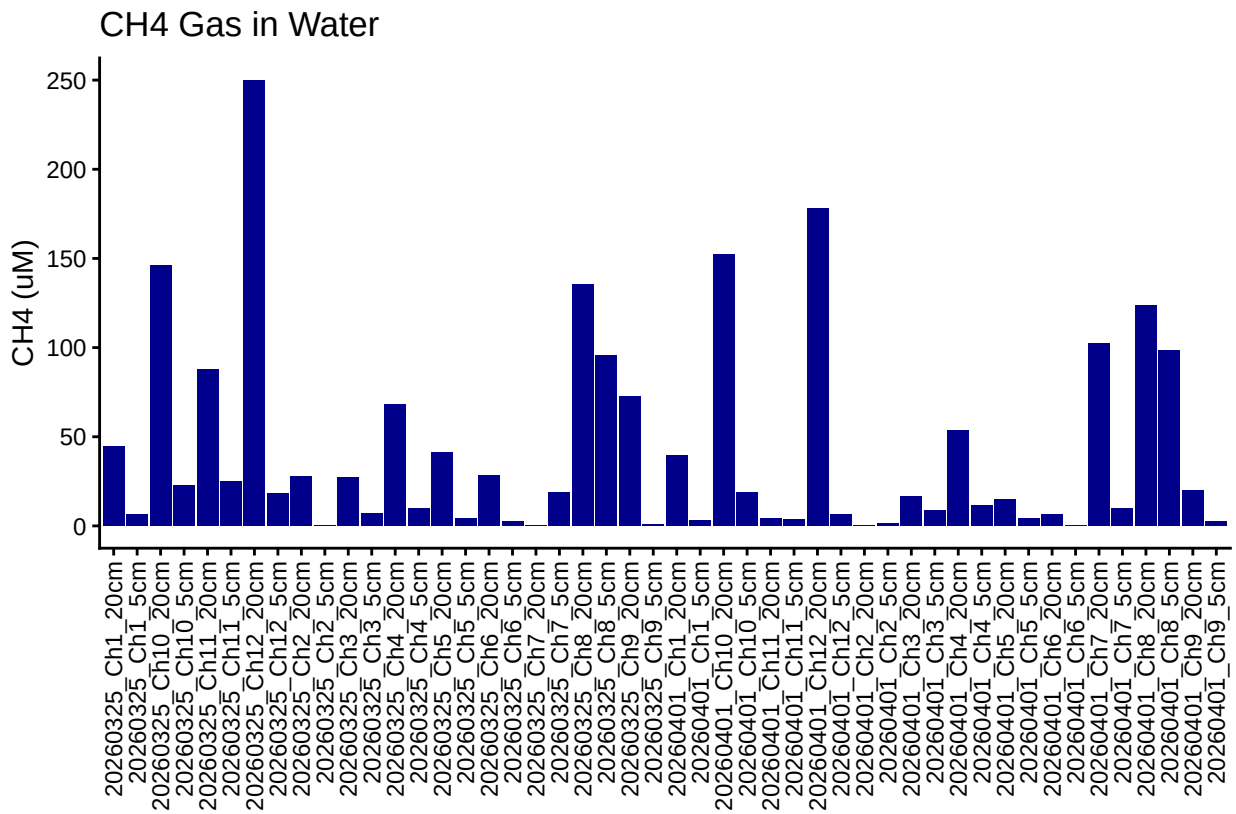
5 Dilution correct samples and Check Range

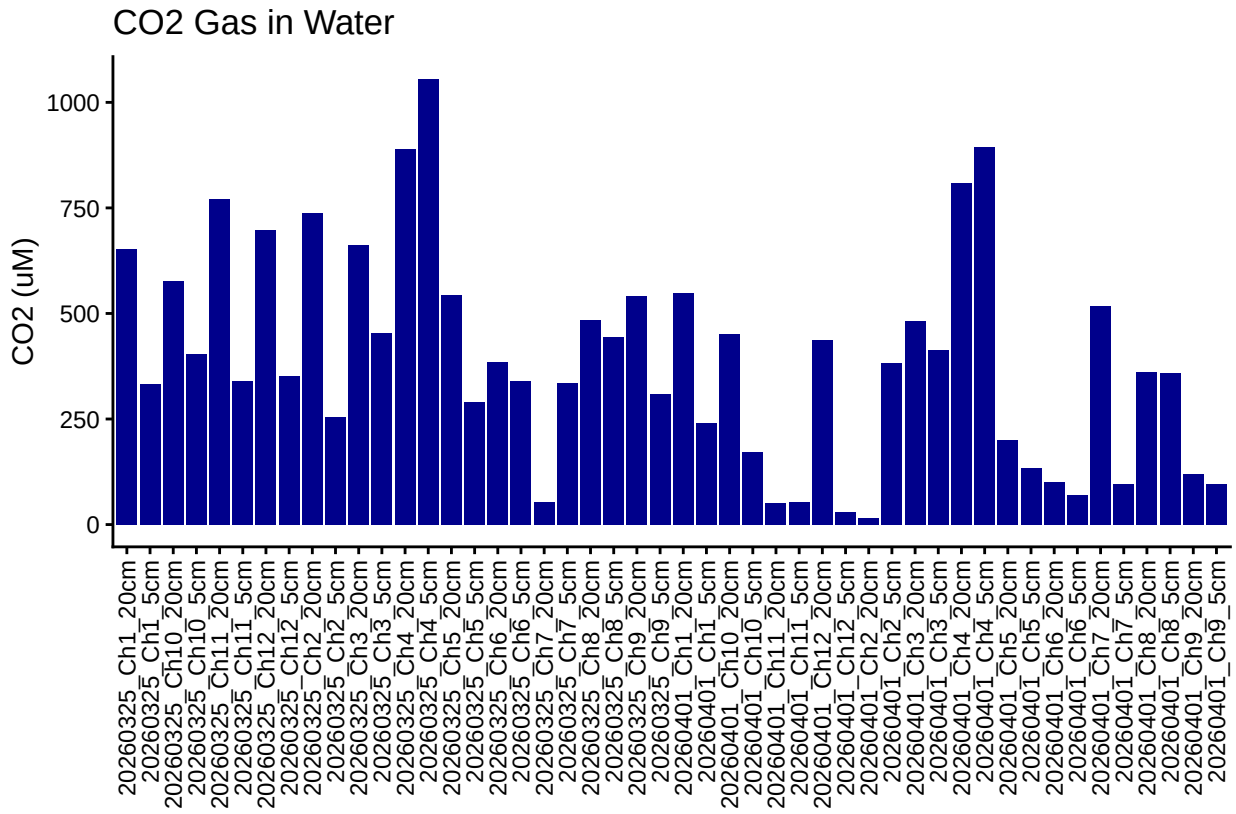


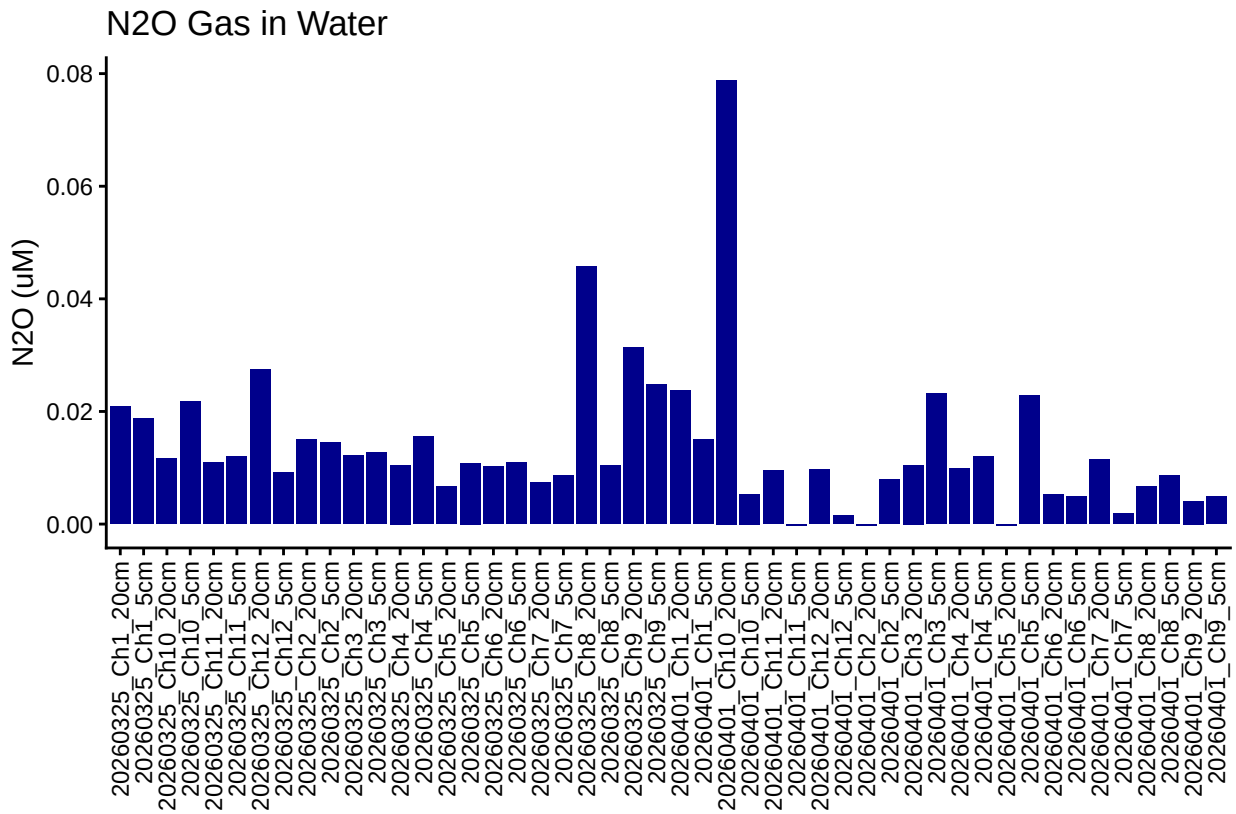



N2O: Green = Within Range (Not Dil Cor)
N2O (ppm)
2.0
1.5
1.0
0.5
0.0
20260325_Ch1_20cm
20260325_Ch1_5cm
20260325_Ch10_20cm
20260325_Ch10_5cm
20260325_Ch11_20cm
20260325_Ch11_5cm
20260325_Ch12_20cm
20260325_Ch12_5cm
20260325_Ch2_20cm
20260325_Ch2_5cm
20260325_Ch3_20cm
20260325_Ch3_5cm
20260325_Ch4_20cm
20260325_Ch4_5cm
20260325_Ch5_20cm
20260325_Ch5_5cm
20260325_Ch6_20cm
20260325_Ch6_5cm
20260325_Ch7_20cm
20260325_Ch7_5cm
20260325_Ch8_20cm
20260325_Ch8_5cm
20260325_Ch9_20cm
20260325_Ch9_5cm
20260401_Ch1_20cm
20260401_Ch1_5cm
20260401_Ch10_20cm
20260401_Ch10_5cm
20260401_Ch11_20cm
20260401_Ch11_5cm
20260401_Ch12_20cm
20260401_Ch12_5cm
20260401_Ch2_20cm
20260401_Ch2_5cm
20260401_Ch3_20cm
20260401_Ch3_5cm
20260401_Ch4_20cm
20260401_Ch4_5cm
20260401_Ch5_20cm
20260401_Ch5_5cm
20260401_Ch6_20cm
20260401_Ch6_5cm
20260401_Ch7_20cm
20260401_Ch7_5cm
20260401_Ch8_20cm
20260401_Ch8_5cm
20260401_Ch9_20cm
20260401_Ch9_5cm

6 Calculate gas in water







7 Merge samples with Metadata & check all samples are present

7.1 Organize Data

7.2 Visualize Data

##Export Data

#end